

HKCEE 1985 Mathematics II

Label 1 $\frac{2}{1+x} - \frac{2}{1-x} - \frac{4x}{x^2-1}$

- A. $\frac{1}{1-x}$
 B. $\frac{1}{1+x}$
 C. $\frac{1-7x}{x^2-1}$
 D. $\frac{1-7x}{1-x^2}$
 E. $\frac{3x+1}{1-x^2}$

Label 2 $\frac{b}{a} - \frac{a}{b} = \frac{a}{1} - \frac{b}{1}$

- A. $a+b$
 B. $a-b$
 C. $-a+b$
 D. $-a-b$
 E. $\frac{1}{a} + \frac{1}{b}$

Label 3 If $\frac{ab}{ka+b} = \frac{1}{k}$, then $b =$

- A. $\frac{a}{a-k}$
 B. $\frac{ka}{k-a}$
 C. $\frac{ka}{1-ka}$
 D. $\frac{k^2a}{a-k}$
 E. $\frac{k^2a}{k-a}$

Figure 3 Density 922

Label 4 $(x+y)^{-1}(x^{-2}-y^{-2}) =$

- A. $\frac{1}{x^3} - \frac{1}{y^3}$
 B. $\frac{1}{x^2y} - \frac{1}{xy^2}$
 C. $\frac{1}{xy^2} - \frac{1}{x^2y}$
 D. $\frac{1}{x^2} - \frac{1}{y^2}$
 E. $\frac{1}{x^2y} + \frac{1}{xy^2}$

Figure 4 Density 779

Label 5 If $a - \sqrt{b^2 + c^2} = d$, then $c =$

- A. $d = a \pm b$
 B. $a - d - b$
 C. $\pm \sqrt{d^2 - a^2 + b^2}$
 D. $\pm \sqrt{a^2 - d^2 - b^2}$
 E. $\pm \sqrt{(a-d)^2 - b^2}$

Figure 5 Density 89

Label 6 The L.C.M. of $2a^2 - 2b^2$ and $a^3 - 2a^2b + ab^2$ is

- A. $a-b$
 B. $(a-b)(a+b)$
 C. $2a(a-b)(a+b)$
 D. $2a(a-b)^2(a+b)$
 E. $2a(a-b)^3(a+b)$

Label 7 Let a and b be constants. If $3x^3 - ax^2 + 5x - 3b$ is divisible by $x + 3$ then $3a + b =$

- A. -32
 B. -22
 C. 22
 D. 32
 E. It cannot be determined

Label 8. $\log_{10}(a^2 - b^2) =$

- A. $\log_{10} a$
- B. $\log_{10} b$
- C. $2\log_{10}(a - b)$
- D. $2\log_{10} a - 2\log_{10} b$
- E. $\log_{10}(a + b) + \log_{10}(a - b)$

Label 9. If α and β are roots of $x^2 + 2x - 4 = 0$, then $2^\alpha 2^\beta =$

- A. $\frac{1}{16}$
- B. $\frac{1}{4}$
- C. 2
- D. 4
- E. 16

Label 10. The second term and the fifth term of a geometric progression are -2 and $40\frac{1}{2}$ respectively. The first term is

- A. $\frac{1}{2}$
- B. 6
- C. 8
- D. 15
- E. 18

Label 11. If $a : b = 1 : 2$ and $b : c = 1 : 3$, then $a + b : b + c =$

- A. $1 : 5$
- B. $2 : 3$
- C. $3 : 4$
- D. $3 : 5$
- E. $3 : 8$

Label 12. A hawker bought 120 apples and the cost was \$90. It was found that $\frac{1}{8}$ of the apples were rotten and could not be sold. He sold the rest at \$1 each. What percentage of the cost was his profit?

- A. $11\frac{1}{9}\%$
- B. $14\frac{2}{7}\%$
- C. $16\frac{2}{3}\%$
- D. $28\frac{4}{7}\%$
- E. $33\frac{1}{3}\%$

Label 13. The marked price of a book is double that of its cost. In a sale, what percentage discount was given if the profit made was 20% of the cost?

- A. 10%
- B. 20%
- C. 30%
- D. 40%
- E. 50%

Label 14. John spends 40 minutes to walk from his home to school. If he increases his walking speed by 2 km/h, then it takes only 30 minutes. What is the distance between John's home and his school?

- A. 1 km
- B. 4 km
- C. 6 km
- D. 8 km
- E. 12 km

Label 15. 60% of the students in a school are boys. 70% of the boys and 40% of the girls wear glasses. If 696 students wear glasses, how many students are there in the school?

- A. 1200

- B. 1050
C. 868
D. 849
E. 800

Label 16

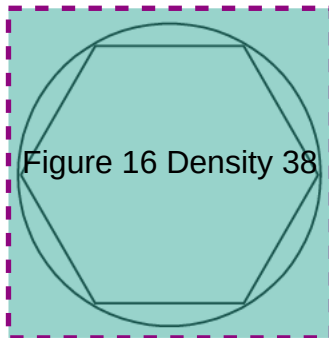


Figure 16 Density 38

In the figure, a regular hexagon of side 2 cm is inscribed in a circle. The area of the circle is greater than the area of the hexagon by

- A. $(3\pi - 6) \text{ cm}^2$
B. $(3\pi - 3\sqrt{3}) \text{ cm}^2$
C. $(4\pi - 6) \text{ cm}^2$
D. $(4\pi - 3\sqrt{3}) \text{ cm}^2$
E. $(4\pi - 6\sqrt{3}) \text{ cm}^2$

Label 17

$\tan \theta \left(\frac{1}{\sin \theta} - \sin \theta \right) =$

- A. 1
B. $\cos \theta$
C. $\sin \theta$
D. $\frac{1}{\cos \theta}$
E. $\frac{1}{\sin \theta}$

Label 18

If $\tan \theta = \frac{2ab}{a^2 - b^2}$ and $0^\circ < \theta < 90^\circ$, then $\cos \theta =$

- A. $\frac{a^2 + b^2}{a^2 - b^2}$
B. $\frac{a^2 - b^2}{a^2 + b^2}$
C. $\frac{a^2 + b^2}{a^2 - b^2}$
D. $\frac{\sqrt{a^2 + b^2}}{a^2 - b^2}$

D
E

Figure 18 Density 496

$$\frac{\sqrt{a^2 - b^2}}{\frac{a^2 + b^2}{\sqrt{a^2 - b^2}}}$$

Label 19

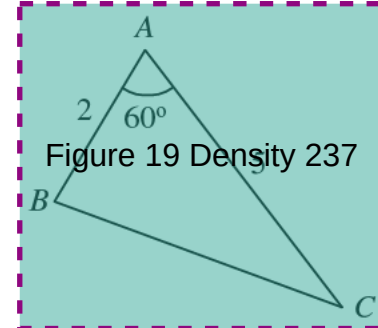


Figure 19 Density 237

In the figure, $AB = 2$ and $AC = 5$, $BC =$

- A. $\sqrt{39}$
B. $\sqrt{29}$
C. $\sqrt{24}$
D. $\sqrt{20}$
E. $\sqrt{19}$

Label 20

In $\triangle ABC$, $\angle A = 30^\circ$, $AB = 6$ cm. If the area of $\triangle ABC$ is 15 cm^2 , $AC =$

- A. 2.5 cm
B. 5 cm
C. 10 cm
D. 12 cm
E. 15 cm

Label 21

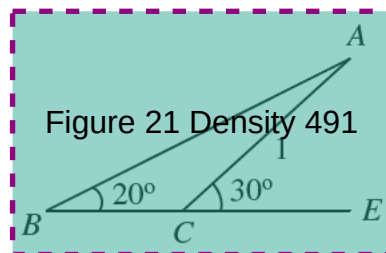


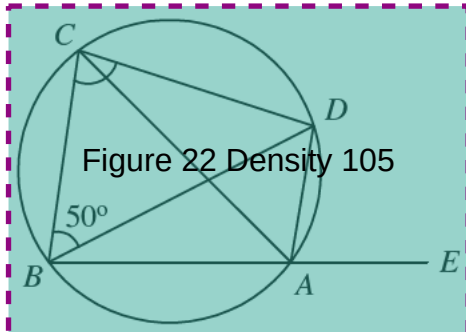
Figure 21 Density 491

In the figure, BCX is a straight line. $AC = 1$, $AB =$

- A. $2 \sin 20^\circ$
B. $2 \cos 20^\circ$
C. $\sqrt{2} \cos 20^\circ$

D. $\frac{1}{2 \sin 20^\circ}$
 E. $\frac{\sqrt{3}}{2 \sin 20^\circ}$

Label 22



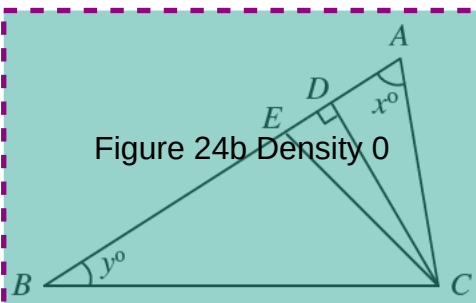
In the figure, $ABCD$ is a cyclic quadrilateral. BA is produced to E . DA bisects $\angle CAE$. $\angle BCD =$

- A. 40°
 B. 45°
 C. 50°
 D. 55°
 E. 65°

Label 23 The exterior angles of a pentagon are $x^\circ, 2x^\circ, 3x^\circ, 4x^\circ, 5x^\circ$. The smallest interior angle of the pentagon is

- A. 120°
 B. 60°
 C. 48°
 D. 36°
 E. 24°

Label 24



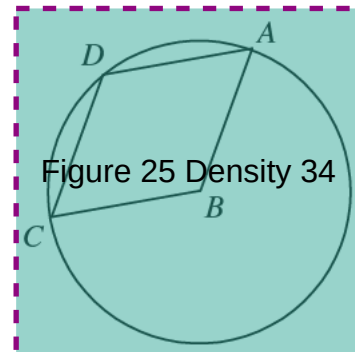
In the figure, A, D, E and B lie on a straight line. CE bisects $\angle ACB$ and $CD \perp AB$. $\angle DCE =$

A. $\frac{1}{2}(x^\circ - y^\circ)$
 B. $\frac{1}{2}(x^\circ + y^\circ)$
 C. $x^\circ - y^\circ$

Figure 24a Density 1193

D. $90^\circ - \frac{1}{2}(x^\circ + y^\circ)$
 E. $90^\circ - (x^\circ - y^\circ)$

Label 25



In the figure, $ABCD$ is a rhombus B is the centre of the circle. $\angle ABC =$

- A. 105°
 B. 120°
 C. 130°
 D. 135°
 E. 150°

Label 26 The distance between $(1 - k, k)$ and $(2, 1 + k)$ is $\sqrt{26}$. $k =$

- A. 4
 B. 6
 C. -4 or 6
 D. 4 or -6
 E. -4 or -6

Label 27 The equation of the perpendicular bisector of the line joining $(1, 2)$ and $(7, 4)$ is

- A. $3x + y + 15 = 0$
 B. $3x + y - 15 = 0$
 C. $3x - y + 9 = 0$
 D. $3x - y - 9 = 0$
 E. $x + 3y - 13 = 0$

Label 28
28

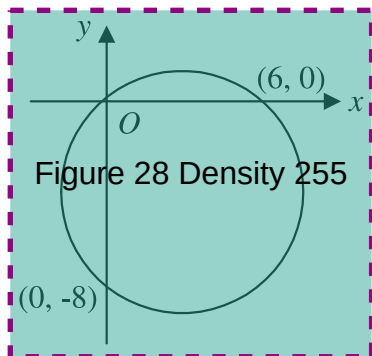


Figure 28 Density 255

In the figure, the circle passes through (0, 0) and cuts the two axes at (6, 0) and (0, -8). Its equation is

- A. $x^2 + y^2 - 3x + 4y = 0$
- B. $x^2 + y^2 + 3x - 4y = 0$
- C. $x^2 + y^2 + 6x - 8y = 0$
- D. $x^2 + y^2 - 6x + 8y = 0$
- E. $x^2 + y^2 - 6x - 8y = 0$

Label 29
29 The equation of a circle is $x^2 + y^2 - 4x - 5 = 0$. Which of the following is/are true?

- I. The circle passes through the origin.
- II. The centre lies on the x-axis.
- III. The line $x - 5 = 0$ touches the circle.

- A. II only
- B. III only
- C. I and II only
- D. II and III only
- E. I, II and III

Label 30
30

Class mid-value	Frequency
$m - 4$	1
m	2
$m + 4$	6

Figure 30 Density 496

The mean of the above distribution is

- A. $m - \frac{1}{3}$

- B. $m - \frac{1}{2}$
- C. $m - 2$
- D. $m - 4$
- E. m

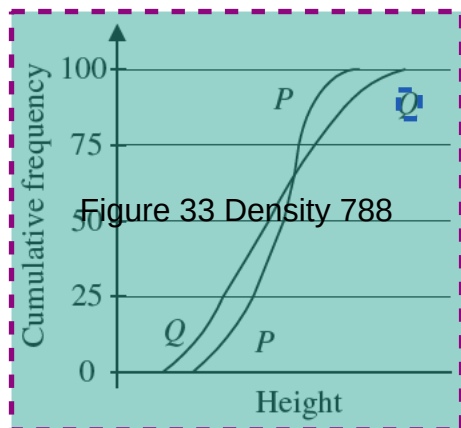
Label 31
31 There are four balls, numbered 1, 2, 5 and 10 in a bag. If 2 balls are taken out at random, the probability that the sum of the numbers on the two balls drawn is greater than or equal to 7 is

- A. $\frac{1}{2}$
- B. $\frac{5}{8}$
- C. $\frac{2}{3}$
- D. $\frac{3}{4}$
- E. $\frac{5}{6}$

Label 32
32 Two dice are thrown. The probability of getting at least one "6" is

- A. $\frac{1}{6}$
- B. $\frac{1}{3}$
- C. $\frac{11}{36}$
- D. $\frac{25}{36}$
- E. $\frac{35}{36}$

Label 33

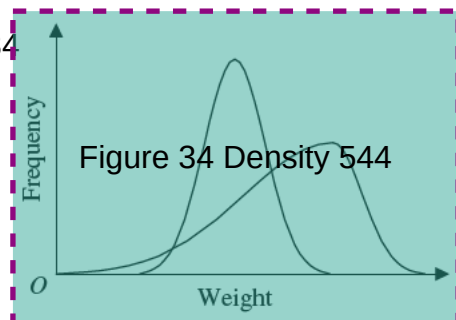


In the figure, P and Q are the cumulative frequency curves for the heights of two groups of students, each having 100 students. Which of the following must be true?

- I. range of $P <$ range of Q
- II. median of $P <$ median of Q
- III. the 3rd quartile of $P <$ the 3rd quartile of Q

- A. I only
- B. II only
- C. I and II only
- D. I and III only
- E. I, II and III

Label 34

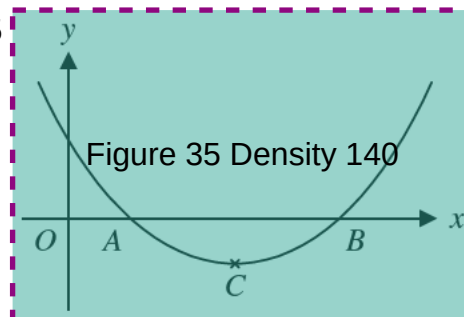


In the figure, P and Q are curves showing the distribution of weights of students in two schools, each having the same number of students. Which of the following must be true?

- I. standard deviation of $P >$ standard deviation of Q
- II. mode of $P >$ mode of Q
- III. median of $P >$ median of Q

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

Label 35



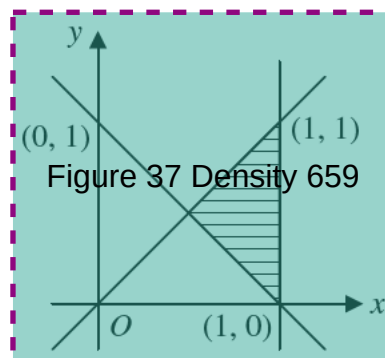
In the figure, the equation of the curve is $y = (x - 2)^2 - 1$. The curve intersects the x -axis at A and B . C is the vertex of the curve. The area of $\triangle ABC$ is

- A. 1
- B. 1.5
- C. 2
- D. 2.5
- E. 3

Label 36 Which of the following is the solution of $(x - 1)(x - 3) \leq 0$ and $x - 2 \leq 0$

- A. $x \leq 2$
- B. $x \leq 3$
- C. $2 \leq x \leq 3$
- D. $1 \leq x \leq 2$
- E. $1 \leq x \leq 3$

Label 37



Which of the following systems of inequalities determine the shaded region in the figure?

- A. $\begin{cases} x \geq 1 \\ x + y \geq 1 \\ x \geq y \end{cases}$
- B. $\begin{cases} x \leq 1 \\ x + y \leq 1 \\ x \geq y \end{cases}$
- C. $\begin{cases} x \leq 1 \\ x + y \geq 1 \\ x \leq y \end{cases}$
- D. $\begin{cases} x \leq 1 \\ x + y \leq 1 \\ x \geq y \end{cases}$
- E. $\begin{cases} x \geq 1 \\ x + y \geq 1 \\ x \geq y \end{cases}$

Figure 37 Density 767

Label 38
38. If a, b, c are in geometric progression, then which of the following is true?

- A. $b^2 = ac$
- B. $b^2 = \frac{ac}{2}$
- C. $b^2 = \frac{a+c}{2}$
- D. $b^2 = \frac{a+c}{2ac}$
- E. $b^2 = \frac{2ac}{a+c}$

Label 39
39. Three distinct numbers x, y and z are in arithmetic progression. Which of the following is/are also in arithmetic progression?

- I. $x + 10, y + 10, z + 10$
- II. $10x, 10y, 10z$
- III. x^2, y^2, z^2

- A. I and II only
- B. I and III only

- C. II and III only
- D. I, II and III
- E. None of I, II and III

Label 40
40. If $f(2x) = 8x^3 + 4x$, then $f(3a) =$

- A. $9a^3 + 6a$
- B. $12a^3 + 6a$
- C. $27a^3 + 6a$
- D. $108a^3 + 6a$
- E. $216a^3 + 12a$

85. A number is first reduced by $p\%$ and then increased by $x\%$. If the number so obtained is the same as the original number then $x =$

- A. p
- B. $\frac{p}{100}$
- C. $\frac{p}{1-p}$
- D. $\frac{100}{100-p}$
- E. $\frac{100p}{100-p}$

Label 42
42. The length and width of a cuboid are each increased by 10% and the height remains unchanged. The percentage increase in volume is

- A. 10%
- B. 20%
- C. 21%
- D. 24%
- E. 33%

Label 43
43. A cone of base radius $2r$ cm and height h cm has a volume of 60 cm^3 . The volume of a cylinder of base radius r cm and height $4h$ cm is

- A. 60 cm^3
- B. 120 cm^3
- C. 180 cm^3
- D. 240 cm^3

E. 360 cm^3

Label 44

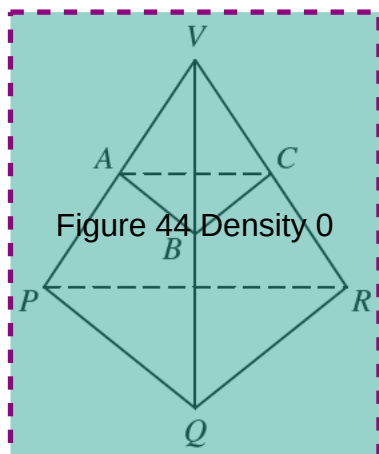


Figure 44 Density 0

In the figure, the volumes of the pyramids $VABC$ and $VPQR$ are 27 cm^3 and 64 cm^3 respectively. Planes ABC and PQR are parallel.
Area of $\triangle ABC$: Area of $\triangle PQR$ =

- A. $\sqrt{27} : \sqrt{64}$
- B. $\sqrt{37} : \sqrt{64}$
- C. $3 : 4$
- D. $9 : 16$
- E. $27 : 64$

Label 45

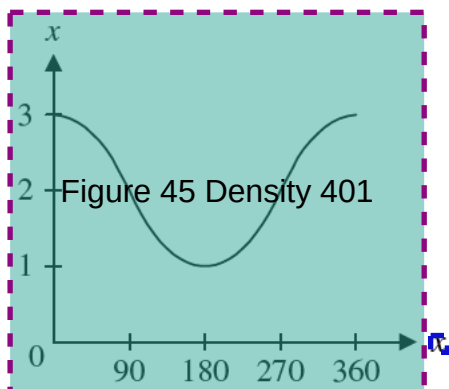


Figure 45 Density 401

The figure shows the graph of

- A. $y = 3\cos x^\circ, 0 \leq x \leq 360$
- B. $y = 3\sin x^\circ, 0 \leq x \leq 360$
- C. $y = 2 + \sin x^\circ, 0 \leq x \leq 360$
- D. $y = 2 + \cos x^\circ, 0 \leq x \leq 360$
- E. $y = 3 + \sin x^\circ, 0 \leq x \leq 360$

Label 46 If $0^\circ \leq \theta \leq 360^\circ$, then the largest value of $2 \sin 2\theta + \cos 2\theta + 2$ is

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Label 47

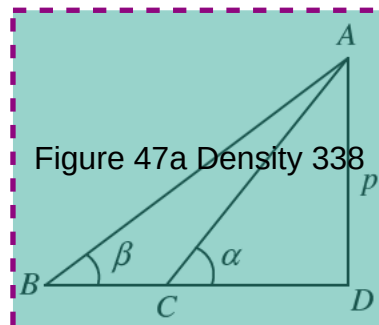


Figure 47a Density 338

In the figure, BCD is a straight line $AD \perp p$, then $BC =$

- A. $p \tan (\beta - \alpha)$
- B. $p (\tan \alpha - \tan \beta)$
- C. $p (\tan \beta - \tan \alpha)$
- D. $p \left(\frac{1}{\tan \alpha} - \frac{1}{\tan \beta} \right)$
- E. $p \left(\frac{1}{\tan \beta} - \frac{1}{\tan \alpha} \right)$

Figure 47b Density 992

Label 48

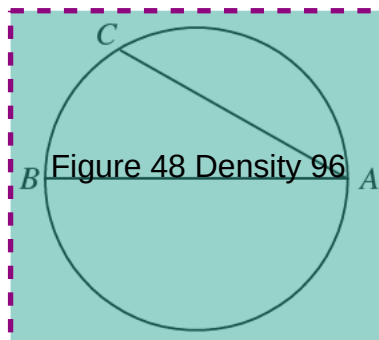


Figure 48 Density 96

In the figure, AB is a diameter of the circle ABC . If arc AC has the same length as AB , then $\angle CAB =$

- A. $\frac{\pi}{2}$ radians
- B. $\left(\frac{\pi}{2} - \frac{1}{2} \right)$ radians
- C. $\left(\frac{\pi}{2} - 1 \right)$ radians

- D. $(\frac{\pi}{2} - 2)$ radians
E. $(\pi - \frac{1}{2})$ radians

Label 49

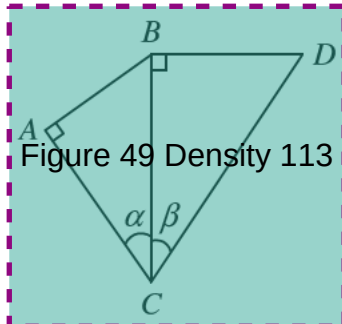


Figure 49 Density 113

In the figure, $\angle CAB = \angle CBD = 90^\circ$. $BC = 2$. The area of quadrilateral $ABCD =$

- A. $2 \sin(\alpha + \beta)$
B. $2(\tan \alpha + \tan \beta)$
C. $2(\sin \alpha \cos \alpha + \sin \beta \cos \beta)$
D. $2(\tan \alpha + \sin \beta \cos \beta)$
E. $2(\sin \alpha \cos \alpha + \tan \beta)$

Label 50

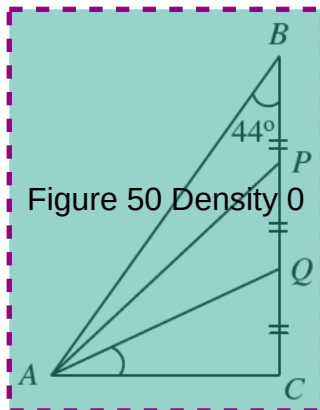


Figure 50 Density 0

In the figure, $\angle C = 90^\circ$. P and Q are points on BC such that $BP = PQ = QC$. $\angle CAQ =$

- A. 30°
B. 25°
C. 22°
D. 20°
E. 15°

Label 51

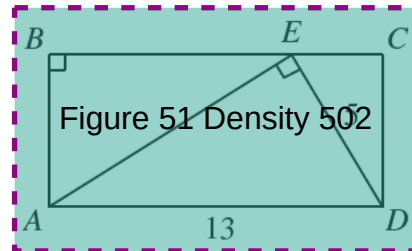


Figure 51 Density 502

In the figure, $ABCD$ is a rectangle. E is a point on BC such that $\angle AED = 90^\circ$. $AD = 13$ and $DE = 5$. The area of $ABCD =$

- A. 30
B. 52
C. 60
D. 65
E. 120

Label 52

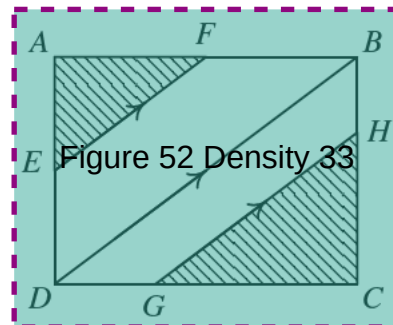
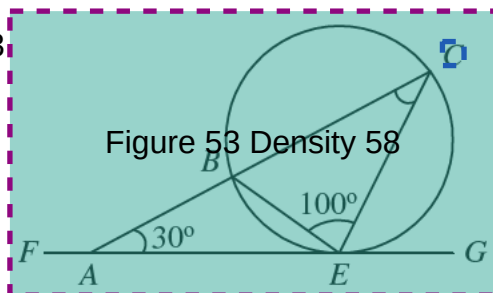


Figure 52 Density 33

In the figure, $ABCD$ is a rectangle E, F, G and H are points on the four sides such that $EF \parallel DB \parallel GH$. $AF = FB$ and $HC = 2BH$. What fraction of the area of $ABCD$ is shaded?

- A. $\frac{13}{36}$
B. $\frac{5}{12}$
C. $\frac{25}{36}$
D. $\frac{25}{72}$
E. $\frac{47}{72}$

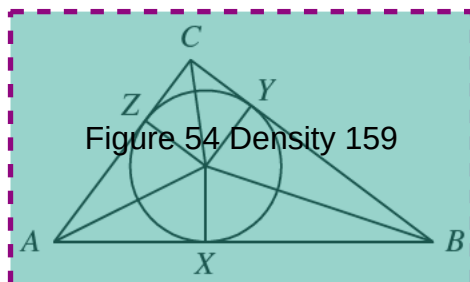
Label 53



In the figure, FG touches the circle at E . The chord CB is produced to meet FG at A . $\angle ACE =$

- A. 10°
- B. 20°
- C. 25°
- D. 30°
- E. 35°

Label 54



In the figure the circle touches the sides of $\triangle ABC$ at X , Y and Z . O is the centre of the circle. Which of the following must be true?

- I. OA bisects $\angle BAC$
- II. A, X, O and Z are concyclic
- III. $AX = AZ$

- A. III only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III